

Operation manual for PA-ARH-100-HVPS

High-voltage power supply for

CF34-mount compact 30kV RHEED system



Pascal Co., Ltd.

Head office

1-16-4 Showa-cho, Abeno-ku, Osaka, 545-0011 Japan

Telephone : +81-6-6626-1321

Facsimile : +81-6-6626-1323

E-mail: pascal@pascal-co-ltd.co.jp

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1 Introduction

1-1 Outline of Pascal RHEED system

High-voltage power supply is individually used for CF34-mount compact Reflection high-energy electron diffraction (RHEED) electron-gun, model PA-ARH-100 series. The power supply can emit 30 kV of acceleration voltage (max.) and 3000 mA of filament current (max.) to drive the hot-cathode electron gun. All e-gun driving parameters can be controlled by remote control box.

Units or parts of the RHEED system are danger. See following chapters and individual documents for assembled parts which show functions of the units and the parts in detail.



1-2 Caution



X-ray is generated when an electron beam is irradiated to a substrate with RHEED (Reflection High Energy Electron Diffraction) gun. While irradiating, be away from other viewing ports than the screen.



Wiring relay terminals and wire connectors wire numerous electric power lines and sensor signal cables. In the case that their covers are removed or the connectors are separated, be sure that the turning the control power supply off.



This system basically requires being electrified continuously. Connecting the system to a power line of insufficient capacity may cause an accidental electric power failure or system malfunction. The rating of electric power line should be sufficient.

2 Construction of RHEED system

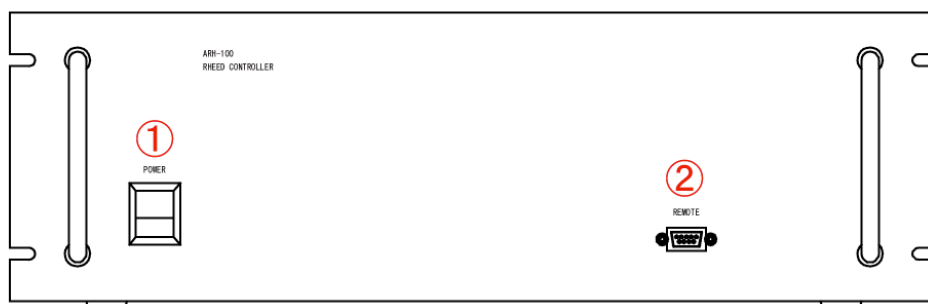
The high-voltage (HV) power supply for PA-ARH-100 series consists following units or items.

- Main body.
- Remote control box.
- Power inlet cable (2 m length).
- Connection cable between the main body and the remote control box (5 m length).
- HV cable (5 m length).
- Deflection driving cable (5 m length).
- Two spare fuses.

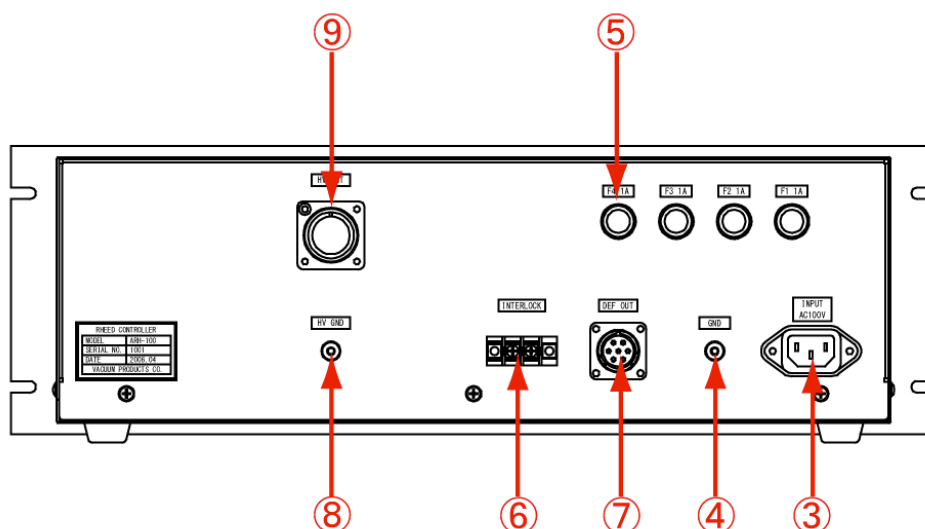
2-1 Main body of high-voltage power supply

Following two figures show panel layout at front side and at rear side of main body of the HV power supply, respectively. Functions of switches, buttons, or plugs in the panels are described below the figures.

(Front side panel)



(Rear side panel)



(1) Main switch

The switch is used to turn on or off main power of the HV power supply. Lamp inside the switch indicates when power on.

(2) Remote controller connector

The female D-sub type plug is the connector to the remote control box via individual connection cable.

(3) Power inlet plug

The plug is the connector to receive AC 100 V of power inlet.

(4) Ground (GND) terminal

This terminal is used to connect earth line to primary side (facility side) for earth of the HV power supply main body.

(5) Fuses

It protects circuit of the HV power supply.

(6) Interlock setting terminal

The signal terminal enables to set interlock setting with relay contact logic.

On the terminal "ON" condition is realized to close the interlock circuit. The ON condition results in high-voltage output and feeding filament current and vice versa.

(7) Deflection output plug

This is the connector of the deflection driver cable to the e-gun.

(8) HV ground (HV GND) terminal

This terminal is used to connect earth line bundled with HV cable to prevent failure operation of the e-gun and electric shock.

(9) HV output plug

The plug connect HV cable to feed acceleration voltage and filament current to the e-gun.

2-2 Remote control box

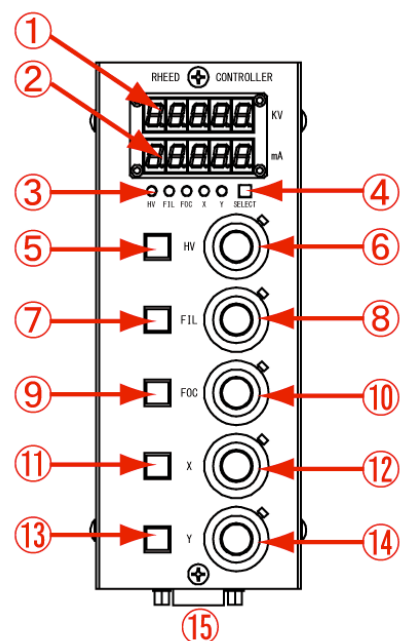
Right figure shows panel layout of the remote control box. Functions of indicators, buttons, or adjuster dials on the panels are described as follows.

(1) High-voltage meter

This numeric (or alphabet) display shows the acceleration voltage when normal operation. Instead of the acceleration voltage error code, which begins a character "E", is shown when the RHEED system alerts. See below section as for the error code.

(2) Current meter

This numeric display exclusively shows e-beam drawing current, cathode current, focusing coil driving current, or X- (or Y-) axis



deflection coil driving current. Display mode is alternately changed by push of selector button (see #3 and #4).

(3) Display mode indicator

This indicator shows which current is shown in the current meter. Illumination at each mode is alternately changed by push of selector button. FIL indicator illuminates at initial state.

(4) Selector button

This button is used to change the display mode on the current meter. One push causes to move next display mode.

(5) HV button

This button is used to turn on/off the HV output. On/off status is alternately changed by one push. Lamp inside the button illuminates when HV turns on.

The RHEED system requires two conditions before turning on the HV output. First it is necessary to turn on the FIL (filament current) and to set the filament current over than a half preset value. (Usually it is enough to turn on FIL with zero position of FIL adjuster.) Next it is necessary to reduce HV value in of HV adjuster less than 0.3.

(6) HV adjuster

The adjuster dial is used to increase or decrease the acceleration voltage of e-gun.

(7) FIL button

This button is used to turn on/off the FIL (cathode current) output. On/off status is alternately changed by one push. Lamp inside the button illuminates when FIL turns on.

(8) FIL adjuster

The adjuster dial is used to increase or decrease the cathode filament current of e-gun.

X, and Y (electron-beam deflection),

(9) FOC button

This button is used to turn on/off the FOC (electron-beam focusing) output. On/off status is alternately changed by one push. Lamp inside the button illuminates when FOC turns on.

(10) FOC adjuster

The adjuster dial is used to increase or decrease the FOC coil driving current of e-gun.

(11) DEF-X button

This button is used to turn on/off the X (electron-beam deflection in horizontal axis) output. On/off status is alternately changed by one push. Lamp inside the button illuminates when X turns on.

(12) DEF-X adjuster

The adjuster dial is used to increase or decrease the X coil driving current of e-gun.

(13) DEF-Y button

This button is used to turn on/off the Y (electron-beam deflection in vertical axis) output. On/off status is alternately changed by one push. Lamp inside the button illuminates when Y turns on.

(14) DEF-Y adjuster

The adjuster dial is used to increase or decrease the Y coil driving current of e-gun.

(15) Cable plug

This plug is used to connect between the remote controller and main body.

3 Installation

3-1 Required condition to set the unit

- (1) No dust, no vibration, and no corrosion atmosphere.
- (2) Environmental temperature from 0 deg C to 30 deg C without exposure to sunshine and environmental humidity less than 90% with dew-condensation free condition.
- (3) Small fluctuation of voltage and frequency of 100 V AC power inlet.
- (4) Enough space for air cooling around the unit.

3-2 Cable setting of power inlet

- (1) Connect AC100V power inlet cable.
- (2) Connect earth cable between GND terminal of the unit and an appropriate ground terminal at power inlet side.

3-3 Cable setting of focusing/deflection coil controller

Connect the coil controller cable at DEF output plug. Order and function of the cable are shown as follows.

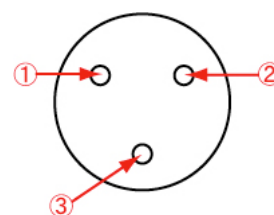
Terminal #	Function	Output polarity
1	DEF-X (horizontal axis e-beam deflection) output	+
2	DEF-X (horizontal axis e-beam deflection) output	-
3	DEF-Y (vertical axis e-beam deflection) output	+
4	DEF-Y (vertical axis e-beam deflection) output	-
5	FOC (focusing e-beam) output	+
6	FOC (focusing e-beam) output	-

3-4 Cable setting between of HV output

- (1) Connect HV cable to HV output plug with notifying pin layout (see right figure).

Terminal #	1	2	3
Function	Filament +	Filament -	HV

- (2) Connect e-gun earth cable to HV GVD terminal. The e-gun earth cable



is bundled with HV cable.

3-5 Interlock setting

Connect an interlock control circuit. The circuit uses contact signal. “OFF” status is realized when the circuit is opened. The “OFF” status causes to halt filament current out and HV out.

4 Operation guide

4-1 Starting

- (1) Turn on the main switch (#1 in sec. 2-1). Confirm the lamp illuminates when power on.
- (2) Confirm the High-voltage meter (#1 in sec. 2-2) shows “0.00”.
- (3) Confirm the current meter (#2 in sec. 2-2) shows “0000”.
- (4) Confirm the display mode indicator (#3 in sec. 2-2) shows “HV” status.

4-2 Initial setting

- (1) Adjust HV, FIL, or FOCUS value to zero in the remote controller.
- (2) Adjust DEF-X or DEF-Y value to 5 in the remote controller.
- (3) Choose display mode to HV or FIL by selector button to observe the filament current first.
- (4) Push FIL button to turn on the filament. Check that the filament current runs around 500 mA after operation.

A interlock function works not to turn on the filament when FIL value is not zero first.

- (5) Push HV button to turn on the acceleration.

A interlock function works not to turn on the acceleration unless the filament current runs. This function effectively works when one mistake the acceleration operation under HV cable disconnection from the electron-gun.

4-3 HV drive for electron beam emission

- (1) Choose display mode to HV by selector button.
- (2) Raise HV value to an appropriate acceleration voltage by HV adjuster. The RHEED e-gun takes warm-up time in a few minutes because of thermal electron emission. So it is necessary to adjust the HV value with considering about 10% HV drop in the warm-up time.

Extraordinary discharge probably occurs when the acceleration operation at initial use of the electron gun or just after differential pumping. Protection function makes HV value zero when discharge current exceeds 250 μ A. Please restart with setting zero values of FIL and HV.

4-4 Filament drive and initial electron-beam alignment

- (1) Choose display mode to FIL by selector button.
- (2) Raise FIL value to the appropriate filament current by FIL adjuster. Electron-beam usually emits when the filament current reaches around 2000 mA.
- (3) Check electron-beam current after the electron-beam emission. The electron-beam current is indicated on the lower line of the display when choosing HV display mode.

First the electron-beam current rapidly increases and then slowly increases. Gradually increase of the electron-beam current is caused by space-charge limited law. And it is the stable operation condition. But less electron-beam current (and the filament current) brings longer filament lifetime.

The electron-beam current occurs to show extraordinary value when the filament is properly set in the Wehnelt-filament assembly. First case is larger current over than 250 μ A. The other case is small current near zero. Reassemble with adjusting filament position when one sees above cases. The electron beam current occurs to show zero due to break of the filament. Exchange the filament in the case.

Extraordinary discharge occurs and makes HV to turn off when the filament unintentionally contacts the Wehnelt. Inform PASCAL about the accident in this case.

- (4) Maximize the electron-beam spot on the fluorescent screen by electron-gun adjuster knob.
Remove obstacles and eliminate external magnetic-field disturbances.
Use the electron-gun adjuster knobs to bring back the electron-beam spot onto the fluorescent screen.

4-5 Focus tuning

- (1) Choose display mode to FOC by selector button.
- (2) Minimize the electron-beam spot on the fluorescent screen by FOCUS adjuster.

4-6 X-Y Deflection setting

- (1) Choose display mode to X (or Y) by selector button.
- (2) Set to an appropriate horizontal (or vertical) position on the fluorescent screen by DEF-X (or DEF-Y) adjuster.

Check the DEF-Y parameter after setting because the vertical position of the electron-beam is important to make the electron-beam path including reflection on the substrate surface.

4-7 Shutdown

- (1) Return the all values to initial state as follows in order.
 - a) DEF-X, DEF-Y: 5 in the adjusters. It is not necessary to return if one wants to restart the

RHEED system as is.

- b) FOCUS: 0 in the adjuster. It is not necessary to return if one wants to restart the RHEED system as is.
- c) FIL: 0 in the adjuster.
- d) HV: 0 in the adjuster.
- (2) Turn off all function buttons in order: DEF-X, DEF-Y, FOCUS, HV, and FIL.
- (3) Turn off the main switch.

5 Troubleshooting / Maintenance

5-1 Alert

The unit shows alerts and halts RHEED operation when it has following problems. Alert codes, which correspond to the problems, is indicated at high-voltage meter

- (1) No signal communication between the main body and remote control box at starting.
Alert code is "E000".
The trouble would be caused by breaking cable or CPU error in main body. First confirm the cable connection for the remote control box.
- (2) "OFF" status in interlock control circuit.
Alert code is "E001".
The trouble would be caused by opening interlock control circuit. Remove factors that bring the circuit-open (ex. lack of interlock terminator).
- (3) Over current of HV output. The alert shows when the HV current exceeds 200 μ A.
Alert code is "E100".
The trouble would be caused by the RHEED operation under bad condition like high-pressure with plenty gas-flow, contamination around the filament and the Wehnelt, or other problems. Change the RHEED operation condition.
- (4) Less filament current. The alert shows when the filament current falls a half of the set value.
Alert code is "E101".
The trouble would be caused by the breakdown of the filament or other problems. Confirm the HV cable setting for the e-gun. One of restoration methods is exchanging filament.

5-2 Daily check

- (1) Fasten the plugs for all kinds of cables.
- (2) Remove the cable damage with respect to snapping, bending, or other troubles.
- (3) Clean around the main body for air cooling. Remove the cap screw to fix the aperture holder inside the electron-gun body.

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High-voltage power supply unit

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Pascal Co., Ltd.

Headoffice 1-16-4 Showa-cho, Abeno-ku, Osaka 545-0011, Japan

TEL : +81-6-6626-1321 FAX : +81-6-6626-1323

Contact-address pascal@pascal-co-ltd.co.jp